

ROBOT GYM

Competitive infrastructure for embodied AI.

A live 1v1 arena where robot policies compete, humans intervene, and every action becomes evaluation evidence.

INVESTOR BRIEF • 2026

KAUSHIK SIVAKUMAR / FOUNDER



ACTUAL PRODUCT ARENA

Robotics demos prove possibility. They rarely prove reliability.

The missing layer is not another model. It is a credible way to test models under pressure.

01

Success hides failure

Polished demos compress retries, recoveries, latency and perception misses into one highlight.

02

Benchmarks lack stakes

Static leaderboards reveal scores—not how policies behave in a live, adversarial scenario.

03

Hardware iteration is costly

Every physical test consumes operator time, robot time and safety margin.

Opportunity: turn evaluation into something rigorous enough for labs—and compelling enough for an audience.

A live arena for robot intelligence.

Same task. Same physics. Different minds.

Policies and people race Unitree G1 robots to acquire, carry and deliver a payload.

AI VS AI

AI VS HUMAN

HUMAN VS HUMAN

LIVE PRODUCT CAPTURE / VLGE x MUJOCO

What the customer gets

● Watch

A shared, browser-viewable match with live state and rationales.

● Control

Gamepad or keyboard input beside model-driven agents.

● Measure

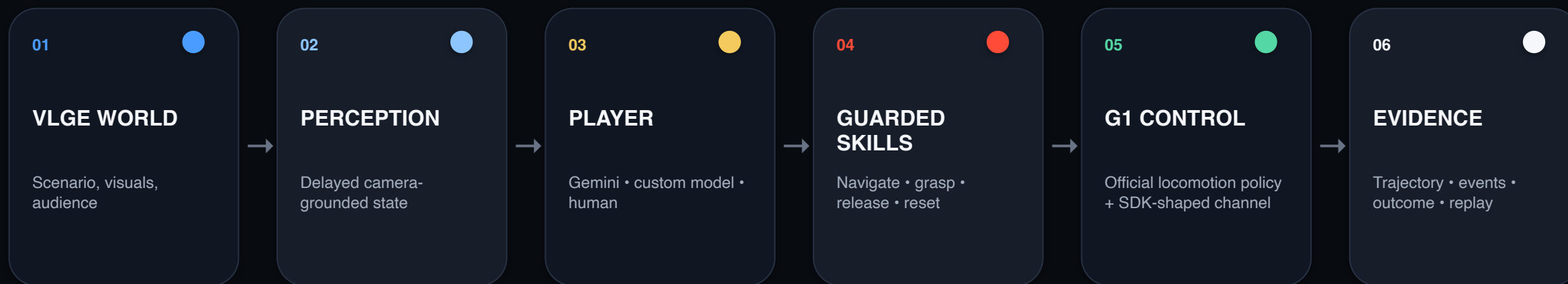
Trajectories, events, commands, falls and recovery attempts.

● Replay

An evidence package for debugging, comparison and trust.

BUILT AND RUNNING

One arena. A clean boundary between reasoning, control and evidence.



KEY DESIGN CHOICE

Models choose grounded intent; the safety and control layer owns executable motion.

This makes model comparison possible without pretending a language model is a motor controller.

Embodied AI is moving from demos to an evaluation economy.

\$38B

PROJECTED HUMANOID ROBOT MARKET BY 2035

Goldman Sachs Research

542K

INDUSTRIAL ROBOTS INSTALLED IN 2024

International Federation of Robotics

+42%

GROWTH IN RAAS FLEETS IN 2024

International Federation of Robotics

MODEL SHIFT

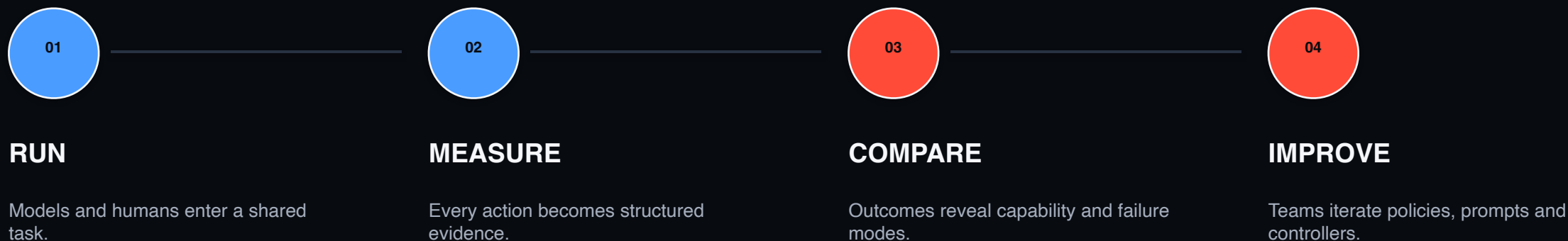
Embodied reasoning is now API-accessible—planning and spatial understanding can be evaluated across models.

Robot Gym is the neutral arena between rapidly changing models and slow, expensive hardware.

Sources: Goldman Sachs Research (2024); IFR World Robotics 2025; Google DeepMind Gemini Robotics-ER.

Competition turns evaluation into a product.

A race is legible to an audience—and unforgiving to a policy.



Flywheel: more scenarios → more matches → better failure data → stronger benchmarks → more teams.

The core arena is already built.

MATCH

- AI vs AI
- AI vs human
- Human vs human
- Rematch + idle player

CONTROL

- Browser gamepad + keyboard
- Official G1 locomotion policy
- 500 Hz physics / 50 Hz channel
- Guarded grasp, release + reset

EVALUATION

- Camera-grounded perception
- Latency, dropout + noise
- Custom HTTP policy adapter
- Evidence and replay package

DEPLOYMENT

Mac development → RunPod GPU host → browser spectators and players

Current product scope; not a claim of real-hardware certification.

Land as evaluation infrastructure. Expand through data and events.

01	PLATFORM	Annual team / enterprise subscription	Private arenas, model adapters, benchmark management and collaboration.
02	COMPUTE	Metered simulation and replay	Hosted matches, accelerated scenario runs and evidence retention.
03	PROGRAMS	Benchmarks, tournaments and services	Custom evaluation suites, partner challenges and failure analysis.

Initial buyer: robotics labs and embodied-AI teams that need repeatable cross-model evaluation.

Start with teams that feel the pain. Earn the audience after the signal.

NOW

DESIGN PARTNERS

Robotics labs + model teams

Install the arena, define one decision-critical benchmark, measure iteration time.

NEXT

PUBLIC CHALLENGE

Developers + research community

Publish a task season, verified match evidence and a model-neutral leaderboard.

SCALE

PRIVATE ARENAS

Enterprises + robot OEMs

Sell proprietary scenarios, access control, fleet evaluation and hosted compute.

North-star metric

Verified evaluation runs per active team

The moat compounds across four layers.

01

SCENARIO GRAPH

A growing library of tasks, resets, constraints and win conditions.

02

FAILURE DATA

Comparable traces of perception, reasoning, control and recovery.

03

CONTROL BOUNDARY

Hardware-shaped channels that make policy claims more honest.

04

COMPETITION NETWORK

Models, teams, spectators and partners improve the benchmark together.

Each match improves the product even when the underlying models change.

From one compelling race to the evaluation layer for embodied AI.

BUILT

G1 DELIVERY ARENA

- Three match modes
- Model + human control
- RunPod deployment
- Evidence capture

12 MONTHS

HOSTED BENCHMARK SUITE

- Accounts + private arenas
- Multi-task scenario library
- Verified leaderboards
- Design-partner pilots

24 MONTHS

HARDWARE-BACKED NETWORK

- Real-unit adapter validation
- Partner robot profiles
- Tournament seasons
- Enterprise evaluation API

Milestones are proposed and subject to customer discovery, funding and hardware partner access.

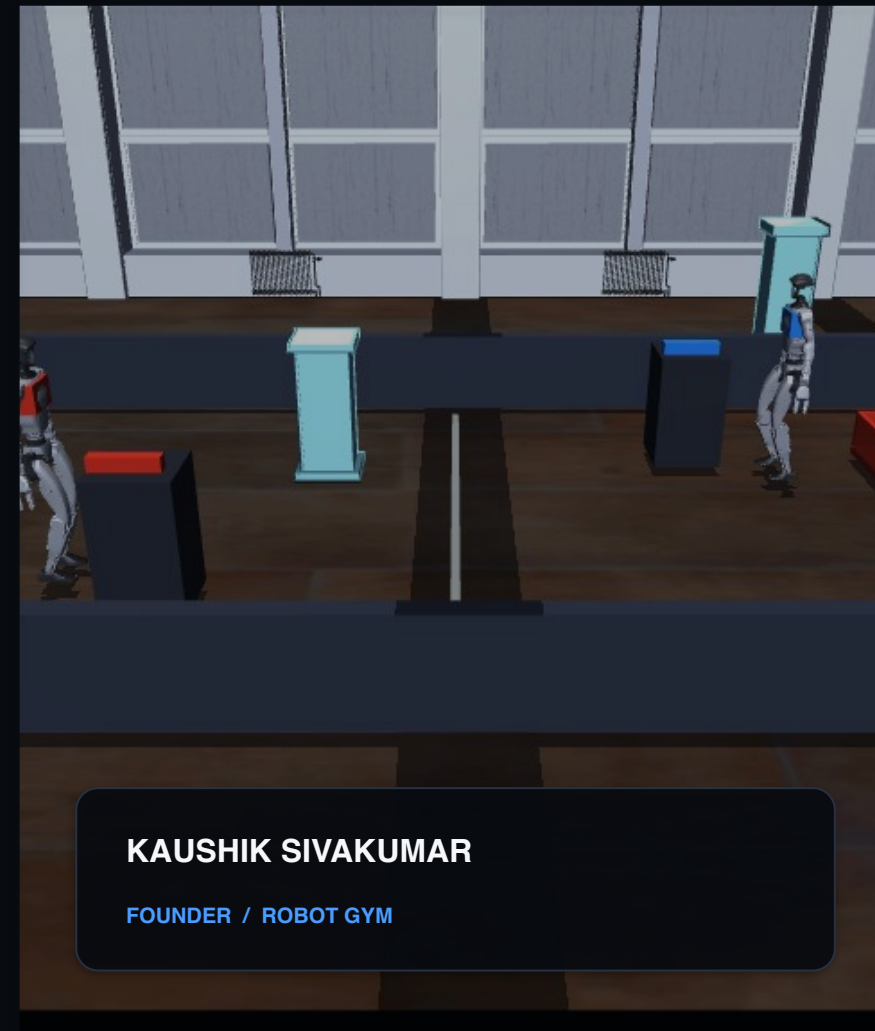
THE ASK

Help turn robot evaluation into a live, trusted market.

Seeking pre-seed capital and design partners.

- | | | |
|----|-----------------|---|
| 01 | PRODUCT | Hosted platform, scenario tooling and evidence UX |
| 02 | ROBOTICS | Benchmark design, controls validation and hardware bridge |
| 03 | MARKET | Design partners, flagship challenge and community |

github.com/KaushikSiva/robot-gym



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LET'S BUILD THE ARENA.